

# Griffin Edwards

Ph.D. Candidate in Mathematics, Georgia Institute of Technology

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## Research Interests

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Algebraic combinatorics, focusing on matroid theory, tropical geometry, algebraic geometry, and pastures.

## Education

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**Georgia Institute of Technology, Atlanta, Georgia**

August 2022–May 2027  
(expected)

Ph.D. in Mathematics

Doctoral Minor in Higher Education and Tech to Teaching Certificate (in progress) – Center for Teaching and Learning

Dissertation (working title): *Dressians of Matroids through Pastures*

Advisor: Matthew Baker

**Brown University, Providence, Rhode Island**

September 2018–May 2022

Bachelor of Science in Mathematics, with Honors

## Teaching

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**Instructor of Record, Georgia Institute of Technology**

Fall 2026

*MATH 1499: Support for Introductory Calculus*

- Taught a flipped, discussion-centered course for 15 students, using interactive readings, collaborative worksheets, and board presentations.
- Implemented an initial diagnostic assessment to identify common prerequisite gaps and incorporated targeted review into recurring class activities.
- Revised course materials to improve screen-reader accessibility and mathematical readability.

**Lecturer's Assistant, Georgia Institute of Technology**

Summer 2026

*MATH 4107: Abstract Algebra I*

**Teaching Assistant, Georgia Institute of Technology**

Fall 2024–Spring 2026

- *MATH 2603: Introduction to Discrete Mathematics* — Fall 2025 and Spring 2026
- *MATH 1552: Integral Calculus* — Spring 2025
- *MATH 1553: Introduction to Linear Algebra* — Fall 2024

**Undergraduate Teaching Assistant, Brown University**

2020–2021

Teaching support for Honors Linear Algebra, Real Analysis, Differential Geometry, Linear Algebra, and Advanced Classical Mechanics.

**Tutor, Sheridan Center for Teaching and Learning, Brown University**

September 2019–May 2021

Tutored students in mathematics, physics, and logic.

## Research Mentoring

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**Research Mentor and Coordinator, Georgia Tech Mathematics REU**

Summer 2025

Co-mentored Catie Lillja and Bora Olmez with Matthew Baker on *Tropical Elliptic Curve Group Law Algorithm*.

**Mentor, Georgia Tech Directed Reading Program**

Fall 2022–Spring 2026

Two students a semester on topics such as Tutte Polynomial, Category Theory, Elliptic Curves, Group Theory in Crystal Structures

## Research & Scholarly Work

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**Manuscripts in preparation**

- *Foundations of Elliptic Matroids*, with Matthew Baker and Noah Solomon. Manuscript in preparation.
- *Foundations of Matroids, Part 3*, with Matthew Baker and Oliver Lorscheid. Manuscript in preparation.

**Software in development**

- *Pastures*, a Macaulay2 package, with Justin Chen and Tianyi Zhang.

- *Rigidity*, a Macaulay2 package, with Kalina Mincheva, Daniel Irving Bernstein, and additional collaborators.

## Selected research contribution

- Computationally verified the conjectured reduced-Dressian behavior of the elliptic matroids  $T_n$  for  $7 \leq n \leq 26$ ; this computation is acknowledged in Baker–Huh–Kummer–Lorscheid, *Lorentzian Polynomials and Matroids over Triangular Hyperfields* (2025).

## Presentations

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| <i>Title to be announced.</i> Tulane University, New Orleans, Louisiana.                                                                                | November 18, 2026 |
| <i>Tropical Elliptic Curves and Matroids.</i> Poster presented at the ISM Discovery School: Combinatorics of Moduli of Curves, LACIM, Montréal, Canada. | May 11–15, 2026   |
| <i>Macaulay2 and Matroids.</i> Computer Algebra Club/Seminar presentation, Georgia Institute of Technology.                                             | November 6, 2025  |
| <i>Algebra from Projective Geometry.</i> Algebra Student Seminar, Georgia Institute of Technology.                                                      | November 17, 2023 |

## Awards & Fellowships

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|--------------------------------------------------------------------------------------------------------------|----------------------|
| MathLab Award for Top TA Helper, Georgia Tech School of Mathematics                                          | 2026                 |
| Recognized for exceptional student support in the department’s communal drop-in office-hour and study space. |                      |
| GAANN Grant, U.S. Department of Education                                                                    | January 2025–present |
| C-PIES GradSTEP Grant, Georgia Tech College of Sciences                                                      | January–May 2025     |
| FESTA Fellowship, Georgia Institute of Technology                                                            | April 2024           |
| David L. Brown Fellowship Award, Georgia Institute of Technology                                             | April 2023           |
| Research to Training Fellowship, Georgia Institute of Technology                                             | August 2022          |
| Undergraduate Teaching and Research Award, Brown University                                                  | Summer 2020          |

## Conferences & Workshops

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|----------------------------------------------------------------------------|--------------|
| Macaulay2 Workshop, Tulane University; contributor                         | April 2025   |
| Tropical Geometry: Moduli Spaces and Matroids, Goethe University Frankfurt | October 2024 |
| Macaulay2, Computational Algebraic Geometry and String Theory; contributor | May 2024     |
| Preliminary Arizona Winter School; participant                             | Fall 2023    |
| Arizona Winter School; participant                                         | March 2023   |
| Connecticut Summer School in Number Theory; participant                    | June 2022    |

## Service, Leadership, & Outreach

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Volunteer and Problem Writer, High School Math Day, Georgia Institute of Technology                                                                                                                 | 2025–2027             |
| College of Sciences Graduate Student Committee Member, Georgia Tech                                                                                                                                 | 2024–present          |
| With C-PIES, select speakers for the <i>Making Science Accessible Seminar Series</i> ; founding organizer of the annual Bowling and Poster Competition, coordinating judges, awards, and publicity. |                       |
| Graduate Peer Mentor Coordinator, Georgia Institute of Technology                                                                                                                                   | 2023–2026             |
| President, Graduate AMS Chapter, Georgia Institute of Technology                                                                                                                                    | 2023–2026             |
| Teacher, Emory Math Circle                                                                                                                                                                          | 2022–2026             |
| Graduate Student Representative, Department of Mathematics, Georgia Institute of Technology                                                                                                         | 2023–2024             |
| Representing the interests of the graduate student in faculty meetings.                                                                                                                             |                       |
| President, Mathematics Department Undergraduate Group, Brown University                                                                                                                             | 2020–2022             |
| Founder and Coordinator, Presentation Theory, Brown University                                                                                                                                      | Fall 2021             |
| Planned and moderated mathematics seminars by undergraduates for undergraduates.                                                                                                                    |                       |
| Directed Reading Program — <i>Rational Points on Elliptic Curves</i> , Brown University                                                                                                             | Fall 2021             |
| Volunteer, Brown University Math Circle                                                                                                                                                             | Fall 2021–Spring 2022 |

## Technical Skills and Interests

**Software and programming:** Macaulay2, Python, and LaTeX.

**Other interests:** Board games, climbing, cooking, and juggling.

Last updated September 2026.